

4(a)	max 3 if any membrane-bound organelles drawn inside cell e.g. nucleus, Golgi body, mitochondrion, lysosome, ER, a large vacuole BUT I vesicle / small vacuoles I detail of cell wall / cell surface membrane around flagellar area four from: cell wall ; <i>must add another line to diagram</i> R cellulose cell wall cell (surface) membrane ; A plasma membrane <i>label line to the inside of cell wall</i> I if outer line labelled as cell membrane DNA ; A bacterial chromosome / nucleoid <i>as label</i> <i>allow one or two circular molecules (circle, loop, ball of string, tangled)</i> R if obviously linear R if label includes, histones / histone proteins / chromatin R if nuclear, envelope / membrane, shown R if nucleolus shown or labelled ribosomes ; R 80S ribosomes <i>shown as, dots / small spheres</i> cytoplasm / cytosol ; plasmid ; <i>smaller than DNA, circular</i> AVP ; e.g. 70S / smaller / 17–20 nm, <i>for ribosome</i> murein / peptidoglycan (for cell wall – allow even if cell wall label not added) pilus /pili <i>drawn as external hair-like structure(s)</i> basal granule <i>at base of flagellum</i> capsule drawn to outside (<i>some do have a capsule</i>)	4
4(b)(i)	0.8 % ; R 0.80 %	1

Question	Answer	Marks
4(b)(ii)	<i>any two from:</i> delay in / no, diagnosis ; delay in / no, treatment / therapy ; A feature of, oral rehydration, treatment /solution A ORT / ORS e.g. lack of supply of ORT unwillingness by health workers to give ORT greater belief in herbal remedies thinking drug treatment is sufficient lack of clean water to make up ORT solution no rehydration programmes I improper treatment / treatment not effective <i>ref. to</i> no / lack of, antibiotic therapy ; e.g. lack of supply of / less efficacious (AW) antibiotics used / less supervision in taking full dose <i>ref. to</i> antibiotic resistant strains ; <i>idea that</i> already have a weak immune system / malnourished ; A may have HIV/AIDS (hence weak immune system)	2
4(c)(i)	<i>any one from:</i> damaged sanitation system / poor sanitation following earthquake ; I poor sanitation unqualified damage to, sewage treatment plants / water purification plants ; AW (contaminated) sewage contaminates drinking water ; I pollution A water becomes contaminated lack of purified drinking water; A lack of, clean / treated / safe / bottled, water AVP ; no, proper / safe, disposal of sewage	1

Question	Answer	Marks
4(c)(ii)	<i>any three from:</i> 1 2011, peak in / highest / AW, number of, cases / countries / countries and cases ; A decrease, increase in 2011, decrease <i>for countries</i> A increase to 2011 and decrease <i>for cases</i> 2 (2010 outbreak in) <i>ref. to</i> Haiti and epidemic (so high number of 2011 cases) ; 3 <i>ref. to</i> spread of disease to countries neighbouring Haiti (hence increase in countries in 2011) ; 4 overall / generally / AW, decrease in number of countries with cases of cholera or 2008 54–56 countries and 2015 41–43 countries, with <i>ref. to</i> decrease ; <i>decrease, peak at 2011 / increase in 2011, decrease is mp1 only</i> 5 suggestion for trend in decrease in number of countries ; e.g. improved infrastructure for sewage / water treatment improved health education to prevent spread providing piped (treated) water 6 2008–2010 number of cases increased and number of countries decreased ; 7 AVP ; e.g. lowest / AW, number of cases in 2013 R if also state that 2013 is lowest for countries similar number of countries in 2008 and 2011 but, approx 3× / much higher / stated values, number of cases in 2011	3